

**BUREAU OF INDIAN STANDARDS**

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*भारतीय मानक मसौदा*

**पारम्परिक औषधियों में प्रयोग हेतु बेनकारा मालाबारिका (लैम.) तिरवेंग.**

**मूल की छाल - विशिष्टि**

*Draft Indian Standard*

***Benkara malabarica* (Lam.) Tirveng. Root bark for use in  
Traditional Medicine – Specification**

ICS 11.120.10

Ayurveda Sectional Committee, AYD 01

**Last Date of Comments:** Please refer covering letter

**FOREWORD**

*(Formal clause would be added later)*

*Benkara malabarica* (Lam.) Tirveng, belonging to the family Rubiaceae, is a thorny, much-branched shrub or small tree, often armed with stout, straight spines that occur in axillary pairs, which is a characteristic feature that helps distinguish it from related genera. The leaves are simple, opposite, and elliptic to obovate in shape, measuring up to about 8 × 4 cm, with a short petiole and a glabrous to slightly pubescent surface. The bark is greyish-brown, and the plant typically bears small, fragrant, white to pale yellow flowers, followed by fleshy, globose fruits when mature.

The species is widely distributed throughout India and Sri Lanka, showing a particular abundance in the forests, scrublands, and hedgerows of peninsular India, where it often grows along forest margins, dry deciduous tracts, and rocky or lateritic terrain up to moderate elevations. Ethnobotanically, the plant holds considerable significance in traditional and folk medicine systems.

*Benkara malabarica* (Lam.) Tirveng is an extra pharmacopoeial plant, currently not included in the official pharmacopoeia. Developing standards for such plants is crucial to ensure their consistent quality, safety, and efficacy. Standardization will also support their responsible use.

In the formulation of this standard significant inputs have been drawn from the Monographs on *Anukta Dravya*, Volume 2 published by Institute for Teaching and Research in Ayurveda, Jamnagar, Gujarat. Additional references include information available in the public domain in print and electronic media including authoritative books.

For the purpose of deciding whether a particular requirement of this standard is complied with the final value, observed, or calculated, expressing the result of a test or analysis shall be rounded off in accordance with IS 2: 2022 ‘Rules for rounding off numerical values (*second revision*)’. The number of significant places retained in the rounded off value should be same as that of the specified value in this standard.

*Draft Indian Standard*

**BENKARA MALABARICA (LAM.) TIRVENG. ROOT BARK FOR USE IN  
TRADITIONAL MEDICINE – SPECIFICATION**

**1 SCOPE**

This standard specifies the requirements and method of sampling and tests for root of *Benkara malabarica* (Lam.) Tirveng which belongs to the rubiaceae family.

**2 REFERENCES**

The standards listed in Annex A contain provisions, which through reference in this text, constitute provisions of this standard. At the time of publication, the editions indicated were valid. All standards are subject to revision and parties to agreements based on this standard are encouraged to investigate the possibility of applying the most recent edition of these standards.

**3 REQUIREMENTS**

**3.1 Description**

**3.1.1 Macroscopic Examination of *Benkara malabarica* (Lam.) Tirveng root bark.**

The dried Root bark of *Benkara malabarica* is hard, 4 cm to 5 cm average length, touch is rough. Outer surface is dark greyish brown and inner surface is dark greyish white in color. Bark is aromatic in smell with short fracture, bitter and astringent taste. Outer surface is shallow cracks in appearance.

**3.1.2 Microscopic Examination of *Benkara malabarica* (Lam.) Tirveng root bark**

Root bark of *Benkara malabarica* Tirveng is outermost angular, tangentially elongated and compactly arranged several layered cork cell, differentiated into outer cork and inner cork. Outer cork cells are filled with tannin content and are lignified. Inner cork cells are slightly filled with brown content. Wide cortex made of parenchyma cells. Most of the parenchyma cells filled with simple and compound starch grain and very few of brown content. Some of Parenchyma cells filled with cluster crystal also. Tanniferous cells are present. Discontinuous patches of several layers of stone cells are largely occupied within the cortical region. Few layer of lignified pericyclic fibers followed after stone cells layer. Some of the secondary phloem are present just beneath the pericyclic fibres, consists sieve elements and fibres. Medullary rays uniseriate, biseriate loaded by simple and compound starch grain and cluster crystals

**3.1.3 Powder Microscopy**

Root bark powder of *Benkara malabarica* Tirveng is creamish light yellow in colour with slightly woody odour, fibrous texture, bitter and astringent taste. Diagnostic characters of its powder shows presence of simple fibres, prismatic and cluster crystal, oil globule, tannin content, cork in surface view, cork in transvers view, stone cells, sclereids, simple and compound starch grains.

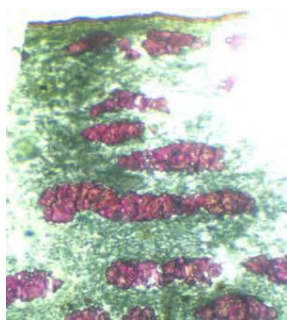


FIG 1A DETAILED  
TRANSVERSE  
SECTION (STAINED)

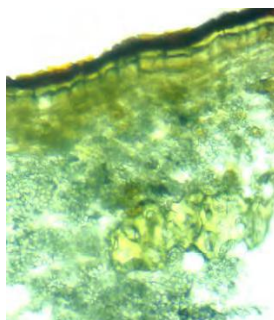


FIG 1B CORK, CORTEX  
WITH STONE CELLS

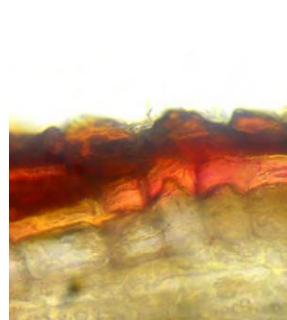


FIG 1C ENLARGED  
VIEW OF CORK CELL

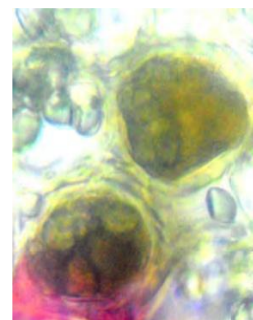


FIG 1D TANNIFEROUS  
CELLS

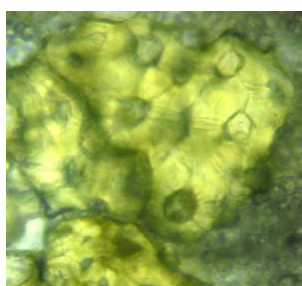


FIG 1E GROUP OF STONE  
CELLS

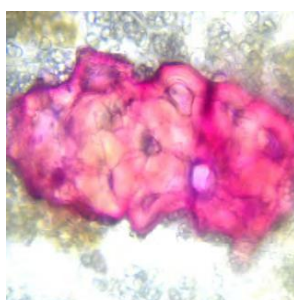


FIG 1F GROUP OF  
STONE CELLS  
(STAINED)

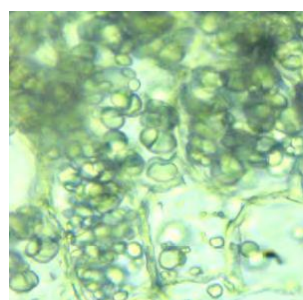


FIG 1G SIMPLE AND  
COMPOUND  
STARCH GRAINS

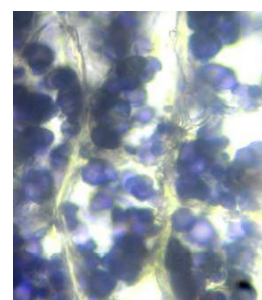


FIG 1H SIMPLE AND  
COMPOUND  
STARCH GRAINS (STAINED)

FIG 1 TRANSVERSE SECTION OF *BENKARA MALABARICA* (LAM.) TIRVENG ROOT BARK



FIG 2A  
ROOT BARK POWDER

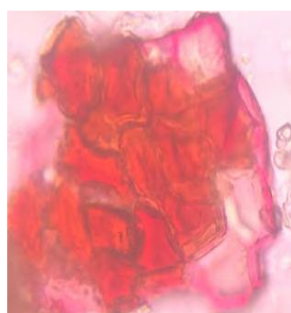


FIG 2B CORK IN  
SURFACE VIEW

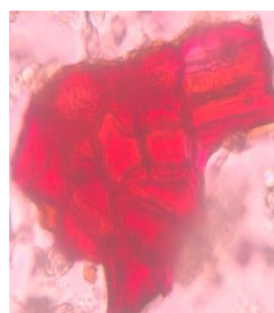


FIG 2C CORK CELLS

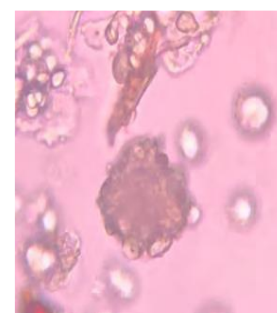


FIG 2D CLUSTER  
CRYSTAL



FIG 2E LIGNIFIED  
SIMPLE FIBERS

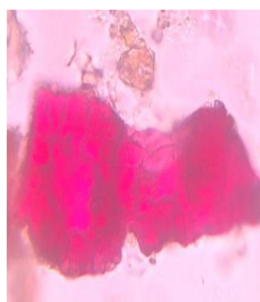


FIG 2F LIGNIFIED  
STONE CELLS

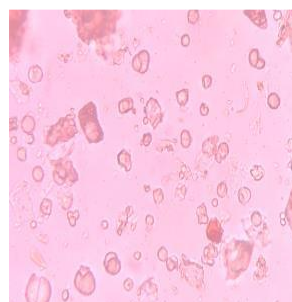


FIG 2G SIMPLE AND  
COMPOUND  
STARCH GRAINS

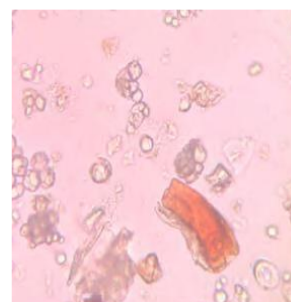


FIG 2H BROWN  
CONTENT

FIG 2 POWDER CHARACTERS OF *BENKARA MALABARICA* (LAM.) TIRVENG. ROOT BARK

### 3.2 General

**3.2.1** *Benkara malabarica* (Lam.) Tirveng root bark shall be free from extraneous/artificial flavors.

**3.2.2** *Benkara malabarica* (Lam.) Tirveng shall comply with physical, chemical and microbiological requirements given in Table 1.

**3.2.3** If any pesticides other than those for which minimum requirements are given in Table 1 are applied to the herb before or after harvesting, those shall also be tested by the methods as specified in IS 17924. Their limit shall be calculated using the following formula:

$$\frac{ADI \times M}{MDD \times 100}$$

where,

*ADI* = acceptable daily intake of pesticide as published by Food and Agriculture Organization-World Health Organization (FAO-WHO), in milligrams per kilogram of body mass;

*M* = body mass in kilograms (60 kg); and

*MDD* = maximum daily dose of the drug, in kilograms.

### 3.3 Quality of Reagents

**3.3.1** Reagents including chemicals used shall be of analytical grade.

**3.3.2** Reagent grade water for laboratory use shall be as per IS 1070.

## 4 SAMPLING

**4.1** Representative samples of the material shall be drawn and tested for conformity to this specification as prescribed in IS 13145.

**4.2** The samples of *Benkara malabarica* (Lam.) Tirveng root bark shall be tested for ascertaining conformity of the material to the requirements in accordance with the relevant clauses given in col (4) of Table 1.

**4.3** The HPTLC analysis shall present the spots or bands specific to the reference standard provided in Annex B.

## 5 PACKING, STORAGE AND MARKING

### 5.1 Packing

*Benkara malabarica* (Lam.) Tirveng root bark shall be packed in clean, sound and dry container made up of metal, glass, food grade polymers, wood or jute bags. The wooden boxes or jute bags shall be suitably coated with moisture proof lining. Visual inspection shall be conducted to ensure that there is no foreign matter within the primary package and it does not impart any foreign smell. Each container shall be securely closed and sealed.

## 5.2 Storage

*Benkara malabarica* (Lam.) Tirveng root bark shall be stored under conditions that prevent contamination and, as far as possible, deterioration. Storage area shall be clean, well ventilated, protected from light, moisture, insects and rodents.

## 5.3 Marking

The following particulars shall be marked or labelled on each pack:

- Name of the material including part of the plant, botanical name, and trade name or brand name, if any;
- Name and address of the producer or packer;
- Batch number;
- Net quantity;
- State and country of production;
- Date of packing (MM/YYYY);
- Instructions for storage; and
- Any other information requested by the buyer, such as the date of harvesting (MM/YYYY) (if known).

The above information, or part of it, may instead appear in the documentation after agreement between the buyer and the seller.

## 5.4 BIS Certification Marking

The product(s) conforming to the requirements of this standard may be certified as per the conformity assessment schemes under the provisions of the *Bureau of Indian Standards Act*, 2016 and the Rules and Regulations framed there under, and the product(s) may be marked with the Standard Mark.

**Table 1 Requirements for *Benkara malabarica* (Lam.) Tirveng Root Bark**  
(Clauses 3.2.2, 3.2.3 and 4.2)

| Sl No. | Characteristic                                                                                 | Requirement | Method of Test, Ref to  |
|--------|------------------------------------------------------------------------------------------------|-------------|-------------------------|
| (1)    | (2)                                                                                            | (3)         | (4)                     |
| i)     | Foreign matter, percent by mass                                                                | Nil         | 6.2 of IS 4333 (Part 1) |
| ii)    | Total ash, percent by mass, <i>Max</i>                                                         | 7.15        | 6 of IS 1797            |
| iii)   | Acid insoluble ash, percent by mass, <i>Max</i>                                                | 2.87        | 8 of IS 1797            |
| iv)    | Alcohol soluble extractive, percent by mass, <i>Min</i>                                        | 24.18       | 10 of IS 1797           |
| v)     | Water soluble extractive, percent by mass, <i>Min</i>                                          | 33.11       | 11 of IS 1797           |
| vi)    | Aflatoxin (B <sub>1</sub> ), ng/kg, <i>Max</i>                                                 | 2.0         | IS 16287                |
| vii)   | Aflatoxin (B <sub>1</sub> +B <sub>2</sub> +G <sub>1</sub> +G <sub>2</sub> ), ng/kg, <i>Max</i> | 5.0         | IS 16287                |
| viii)  | Lead, mg/kg, <i>Max</i>                                                                        | 10.0        | IS 16913                |
| ix)    | Mercury, mg/kg, <i>Max</i>                                                                     | 1.0         | IS 16913                |
| x)     | Cadmium, mg/kg, <i>Max</i>                                                                     | 0.3         | IS 16913                |
| xi)    | Arsenic, mg/kg, <i>Max</i>                                                                     | 3.0         | IS 16913                |
| xii)   | <i>Staphylococcus aureus</i> , per g                                                           | Absent      | Annex C                 |

|        |                                                                           |                                                                                                                               |          |
|--------|---------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|----------|
|        |                                                                           | for extract and powder                                                                                                        |          |
|        |                                                                           | Absent                                                                                                                        |          |
|        |                                                                           | for plant material                                                                                                            |          |
| xiii)  | <i>Pseudomonas aeruginosa</i> , per g                                     | Absent                                                                                                                        | Annex C  |
|        |                                                                           | for extract and powder                                                                                                        |          |
|        |                                                                           | Absent                                                                                                                        |          |
|        |                                                                           | for plant material                                                                                                            |          |
| xiv)   | <i>Salmonella</i> spp., per g                                             | Absent                                                                                                                        | Annex C  |
| xv)    | <i>Escherichia coli</i> , per g                                           | Absent                                                                                                                        | Annex C  |
|        |                                                                           | for extract and powder                                                                                                        |          |
| xvi)   | Total microbial plate count, per g, <i>Max</i>                            | 10 for plant<br>10 <sup>5</sup> for extract and powder<br>10 <sup>7</sup> for plant material which will be treated before use | Annex C  |
| xvii)  | Total yeast and mould, per g, <i>Max</i>                                  | 10 <sup>3</sup> for extract and powder<br>10 <sup>5</sup> for plant material which will be treated before use                 | Annex C  |
| xviii) | Pesticide residues                                                        |                                                                                                                               |          |
| a)     | Alachlor, mg/kg, <i>Max</i>                                               | 0.02                                                                                                                          | IS 17924 |
| b)     | Aldrin and dieldrin (sum of ),mg/kg, <i>Max</i>                           | 0.05                                                                                                                          |          |
| c)     | Azinphos-methyl, mg/kg, <i>Max</i>                                        | 1.0                                                                                                                           |          |
| d)     | Bromopropylate, mg/kg, <i>Max</i>                                         | 3.0                                                                                                                           |          |
| e)     | Chlordane (sum of cis-, trans - and Oxythlordane), mg/kg, <i>Max</i>      | 0.05                                                                                                                          |          |
| f)     | Chlorfenvinphos, mg/kg, <i>Max</i>                                        | 0.5                                                                                                                           |          |
| g)     | Chlorpyrifos, mg/kg, <i>Max</i>                                           | 0.2                                                                                                                           |          |
| h)     | Chlorpyrifos-methyl, mg/kg, <i>Max</i>                                    | 0.1                                                                                                                           |          |
| j)     | Cypermethrin (and isomers), mg/kg, <i>Max</i>                             | 1.0                                                                                                                           |          |
| k)     | DDT (sum of p,p'-DDT, o,p'-DDT, p,p'-DDE and p,p'-TDE), mg/kg, <i>Max</i> | 1.0                                                                                                                           |          |
| m)     | Deltamethrin, mg/kg, <i>Max</i>                                           | 0.5                                                                                                                           |          |
| n)     | Diazinon, mg/kg, <i>Max</i>                                               | 0.5                                                                                                                           |          |
| p)     | Dichlorvos, mg/kg, <i>Max</i>                                             | 1.0                                                                                                                           |          |
| q)     | Dithiocarbamates (as CS <sub>2</sub> ), mg/kg, <i>Max</i>                 | 2.0                                                                                                                           |          |

|     |                                                                                                             |      |          |
|-----|-------------------------------------------------------------------------------------------------------------|------|----------|
| r)  | Endosulfan (sum of isomers and endosulfan sulphate), mg/kg, <i>Max</i>                                      | 3.0  | IS 17924 |
| s)  | Endrin, mg/kg, <i>Max</i>                                                                                   | 0.05 |          |
| t)  | Ethion, mg/kg, <i>Max</i>                                                                                   | 2.0  |          |
| u)  | Fenitrothion, mg/kg, <i>Max</i>                                                                             | 0.5  |          |
| v)  | Fenvalerate, mg/kg, <i>Max</i>                                                                              | 1.5  |          |
| w)  | Fonofos, <i>Max</i>                                                                                         | 0.05 |          |
| y)  | Heptachlor (sum of heptachlor and heptachlor epoxide), mg/kg, <i>Max</i>                                    | 0.05 |          |
| z)  | Hexachlorobenzene, mg/kg, <i>Max</i>                                                                        | 0.1  |          |
| aa) | Hexachlorocyclohexane isomers (other than $\gamma$ ), mg/kg, <i>Max</i>                                     | 0.3  |          |
| bb) | Lindane ( $\gamma$ - hexachlorocyclohexane), mg/kg, <i>Max</i>                                              | 0.6  |          |
| cc) | Malathion, mg/kg, <i>Max</i>                                                                                | 1.0  |          |
| dd) | Methidathion, mg/kg, <i>Max</i>                                                                             | 0.2  |          |
| ee) | Parathion, mg/kg, <i>Max</i>                                                                                | 0.5  |          |
| ff) | Parathion-methyl, mg/kg, <i>Max</i>                                                                         | 0.2  |          |
| gg) | Permethrin, mg/kg, <i>Max</i>                                                                               | 1.0  |          |
| hh) | Phosalone, mg/kg, <i>Max</i>                                                                                | 0.1  |          |
| jj) | Piperonylbutoxide, mg/kg, <i>Max</i>                                                                        | 3.0  |          |
| kk) | Pirimiphos-methyl, mg/kg, <i>Max</i>                                                                        | 4.0  |          |
| mm) | Pyrethrins (sum of), mg/kg, <i>Max</i>                                                                      | 3.0  |          |
| nn) | Quintozene (sum of quintozene, pentachloroaniline and methyl pentachlorophenyl sulphide), mg/kg, <i>Max</i> | 1.0  |          |

ANNEX A

(Clause 2)

LIST OF REFERRED STANDARDS

| <i>IS No.</i>                       | <i>Title</i>                                                                                                                                                                                                                                            |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IS 1070: 2023                       | Reagent grade water – Specification ( <i>fourth revision</i> )                                                                                                                                                                                          |
| IS 1797: 2017                       | Spices and condiments – Methods of test ( <i>third revision</i> )                                                                                                                                                                                       |
| IS 4333: 2018                       | Methods of analysis for food grains: Part 1 Refractions ( <i>third revision</i> )                                                                                                                                                                       |
| IS 13145: 2014                      | Spices and condiments – Methods of sampling ( <i>second revision</i> )                                                                                                                                                                                  |
| IS 16287: 2015/<br>ISO 16050 : 2003 | Foodstuffs – Determination of aflatoxin B <sub>1</sub> , and the total content of aflatoxins B <sub>1</sub> , B <sub>2</sub> , G <sub>1</sub> and G <sub>2</sub> in cereals, nuts and derived products – High performance liquid chromatographic method |
| IS 16913: 2018                      | Methods of test for cosmetics – Determination of heavy metals (Arsenic, Cadmium, Lead and Mercury) by atomic absorption spectrometry (AAS)                                                                                                              |
| IS 17924: 2022                      | Determination of pesticide residue in herbal materials                                                                                                                                                                                                  |

**ANNEX B**

*(Clause 4.3)*

**HIGH PERFORMANCE THIN-LAYER CHROMATOGRAPHY (HPTLC)  
IDENTIFICATION**

**B-1 HPTLC**

**B-1.1 Apparatus**

**B-1.1.1** *Precoated silica gel 60 F254 plate*

**B-1.1.2** *Development chamber*

**B-1.1.3** *HPTLC Scanner*

**B-1.1.4** *Deuterium lamp*

**B-1.1.5** *Mercury lamp*

**B-1.1.6** *Software for analysis*

**B-1.1.7** *Hot air oven*

**B-1.1.8** *Syringe (100µl)*

**B-1.2 Reagents**

**B-1.2.1** *Methanolic extract of test solution*

**B-1.2.2** *Toluene — AR or equivalent grade*

**B-1.2.3** *Ethyl Acetate — AR or equivalent grade*

**B-1.2.4** *Sulphuric Acid — AR or equivalent grade*

**B-1.3 Solvent System**

Dissolve toluene and ethyl acetate in the ratio (9: 1) v/v.

**B-1.4 Procedure**

Apply 5 µl of test solution as 8 mm band at a height of 10 mm from the base of a 20 cm x 10 cm HPTLC plate. Develop the plate to a distance of 8 cm from the band using solvent system. Dry the plate and examine under UV 254 and UV 366 nm. Spray the plate with 4% H<sub>2</sub>SO<sub>4</sub> and heat at temperature of 105 °C to 110 °C till the bands appear. The chromatogram recorded with the test solution is shown in Fig. 3.

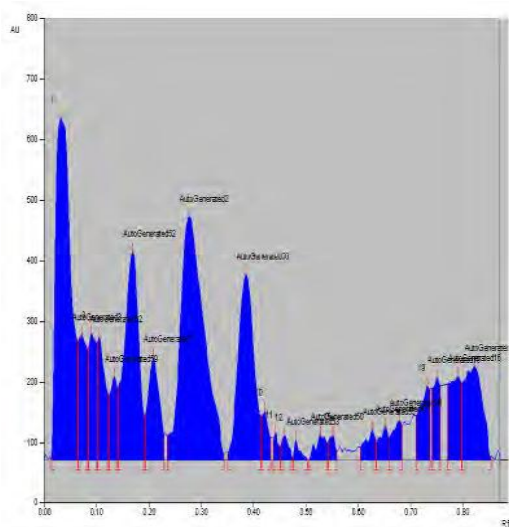


FIG 3A DENSITOGAM OF ROOT  
SAMPLE AT 254 NM

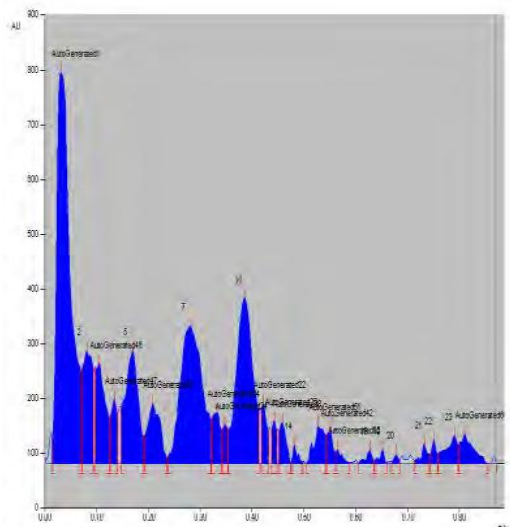


FIG 3B DENSITOGAM OF ROOT  
SAMPLE AT 366 NM

FIG. 3 DENSITOGAM OF ROOT SAMPLE

**ANNEX C**

[Table 1, Sl No. (xii, xiii, xiv, xv, xvi, xvii), Col 4]

**MICROBIAL LIMIT TESTS**

**C-1 GENERAL**

The tests are designed for the estimation of the number of viable aerobic microorganisms present and for detecting the presence of designated microbial species in the extract. The term 'growth' is used to designate the presence and presumed proliferation of viable microorganisms.

**C-2 APPARATUS**

**C-2.1 Oven for Dry Sterilization**

**C-2.2 Autoclave for Wet Sterilization**

**C-2.3 Incubator**

**C-2.4 Water Bath**

**C-2.5 pH-meter**

**C-2.6 Sterile Membrane Filters, 50 mm in Diameter**

**C-2.7 Colony Counting Equipment**

**C-2.8 Analytical Balance**

**C-2.9 Standard Glassware**

**C-3 MEDIA**

Culture media may be prepared as given below or dehydrated culture media may be used provided that, when reconstituted as directed by the manufacturer, they have similar ingredients and / or yield media comparable to those obtained from the formulae given below.

Where agar is specified in a formula, use agar that has a moisture content of not more than 15 percent. Where water is called for in a formula, use purified water. Unless otherwise indicated, the media should be sterilized by heating in an autoclave (15 psi) at 121 °C for 15 min. In preparing media by the formulas given below, dissolve the soluble solids in the water, using heat if necessary, to effect complete solution, add solutions of 0.1N hydrochloric acid or 0.1N sodium hydroxide in quantities sufficient to yield the required pH in the medium when it is ready for use. Determine the pH at 25 °C ± 2 °C.

**C-3.1 Baird Parker Agar Medium**

**C-3.1.1 Composition**

Pancreatic digest of casein     10.0 g

|                  |          |
|------------------|----------|
| Beef extract     | 5.0 g    |
| Yeast extract    | 1.0 g    |
| Lithium chloride | 5.0 g    |
| Agar             | 20.0 g   |
| Glycine          | 12.0 g   |
| Sodium pyruvate  | 10.0 g   |
| Water            | 1 000 ml |

#### **C-3.1.2 Preparation**

Suspend the components in 1 000 ml of water, heat with frequent agitation and boil for 1 min. Sterilize, cool in between 45 °C to 50 °C, add 10 ml of a one percent w/v solution of sterile potassium tellurite and 50 ml of egg yolk emulsion. Mix thoroughly, but gently and pour into plates. (Prepare the egg yolk emulsion by disinfecting the surface of whole shell eggs, aseptically cracking the eggs, and separating out intact yolks into a sterile graduated cylinder. Add sterile saline solution, get a 3 to 7 ratio of egg yolk to saline. Add to a sterile blender cup and mix at high speed for 5 s). Adjust the pH after sterilization to  $6.8 \pm 0.2$ .

### **C-3.2 Bismuth Sulphite Agar Medium**

#### **C-3.2.1 Composition**

##### **Solution (1)**

|                 |          |
|-----------------|----------|
| Beef extract    | 6 g      |
| Peptone         | 10 g     |
| Agar            | 24 g     |
| Ferric citrate  | 0.4 g    |
| Brilliant green | 10 mg    |
| Water           | 1 000 ml |

##### **Solution (2)**

|                                       |        |
|---------------------------------------|--------|
| Ammonium bismuth citrate              | 3 g    |
| Sodium sulphite                       | 10 g   |
| Anhydrous disodium hydrogen phosphate | 5 g    |
| Dextrose monohydrate                  | 5 g    |
| Water                                 | 100 ml |

#### **C-3.2.2 Preparation**

Suspend the components of Solution 1 in 1 000 ml of water. Heat to boiling to dissolve the medium completely. Sterilize by maintaining at 115 °C for 30 min.

Suspend the components of Solution 2 in 100 ml of water. Heat to boiling to dissolve the medium completely. Do not autoclave.

Add 1 volume of solution 2 to 10 volumes of solution 1 which was previously melted and cooled to a temperature of 55 °C. Bismuth Sulphite Agar Medium should be stored at 2 °C to 8 °C for 5 days before use.

### **C-3.3 Brilliant Green Agar Medium**

#### **C-3.3.1 Composition**

|                 |          |
|-----------------|----------|
| Peptone         | 10.0 g   |
| Yeast extract   | 3.0 g    |
| Lactose         | 10.0 g   |
| Sucrose         | 10.0 g   |
| Sodium chloride | 5.0 g    |
| Phenol red      | 80.0 g   |
| Brilliant green | 12.5 mg  |
| Agar            | 12.0 g   |
| Water           | 1 000 ml |

#### **C-3.3.2 Preparation**

Mix the components, allow to stand for 15 min, sterilize by maintaining at 115 °C for 30 min and mix before pouring.

### **C-3.4 Buffered Sodium Chloride Peptone Solution pH 7.0**

#### **C-3.4.1 Composition**

|                                |          |
|--------------------------------|----------|
| Potassium dihydrogen phosphate | 3.56 g   |
| Disodium hydrogen phosphate    | 7.23 g   |
| Sodium chloride                | 4.30 g   |
| Peptone (meat or casein)       | 1.0 g    |
| Water                          | 1 000 ml |

#### **C-3.4.2 Preparation**

Mix the components and heat if necessary to dissolve the medium completely. 0.1 to 1.0 percent w/v Polysorbate 20 or polysorbate 80 may be added. Sterilize by heating in an autoclave at 121°C for 15 min.

### **C-3.5 Casein Soyabean Digest Agar Medium**

#### **C-3.5.1 Composition**

|                                |          |
|--------------------------------|----------|
| Pancreatic digest of casein    | 15.0 g   |
| Papaic digest of soyabean meal | 5.0 g    |
| Sodium chloride                | 5.0 g    |
| Agar                           | 15.0 g   |
| Water                          | 1 000 ml |

#### **C-3.5.2 Preparation**

Suspend the components in water. Heat to boiling to dissolve the medium completely. Sterilize at 121 °C for 15 min in an autoclave and adjust the pH after sterilization to  $7.3 \pm 0.2$ .

### **C-3.6 Cetrimide Agar Medium**

#### **C-3.6.1 Composition**

|                              |          |
|------------------------------|----------|
| Pancreatic digest of gelatin | 20.0 g   |
| Magnesium chloride           | 1.4 g    |
| Potassium sulphate           | 10.0 g   |
| Cetrimide                    | 0.3 g    |
| Agar                         | 13.6 g   |
| Glycerin                     | 10.0 g   |
| Water                        | 1 000 ml |

#### **C-3.6.2 Preparation**

Dissolve the components in 1 000 ml of water. Heat to boiling for 1 min with shaking. Sterilize at 121 °C for 15 min in an autoclave and adjust the pH after sterilization to 7.0 to 7.4.

### **C-3.7 Deoxycholate Citrate Agar Medium**

#### **C-3.7.1 Composition**

|                     |          |
|---------------------|----------|
| Beef extract        | 5.0 g    |
| Peptone             | 5.0 g    |
| Lactose             | 10.0 g   |
| Trisodium citrate   | 8.5 g    |
| Sodium thiosulphate | 5.4 g    |
| Ferric citrate      | 1.0 g    |
| Sodium deoxycholate | 5.0 g    |
| Neutral red         | 0.02 g   |
| Agar                | 12.0 g   |
| Water               | 1 000 ml |

#### **C-3.7.2 Preparation**

Mix the components and allow to stand for 15 min. Gently boil with continuous stirring and continue boiling until solution is complete. Cool to 80 °C, mix, pour and cool rapidly.

Care should be taken not to overheat Deoxycholate Citrate Agar during preparation. It should not be re-melted and the surface of the plates should be dried before use.

### **C-3.8 Fluid Casein Digest Soya Lecithin Polysorbate 20 Medium**

#### **C-3.8.1 Composition**

|                             |          |
|-----------------------------|----------|
| Pancreatic digest of casein | 20 g     |
| Soya lecithin               | 5 g      |
| Polysorbate 20              | 40 ml    |
| Water                       | 1 000 ml |

### **C-3.8.2 Preparation**

Dissolve the pancreatic digest of casein and soya lecithin in water, heating in a water bath at 48 °C to 50 °C for about 30 min to effect solution. Add polysorbate 20, mix and dispense as desired. Sterilize at 121 °C for 15 min in an autoclave.

## **C-3.9 Fluid Lactose Medium**

### **C-3.9.1 Composition**

|                              |          |
|------------------------------|----------|
| Beef extract                 | 3.0 g    |
| Pancreatic digest of gelatin | 5.0 g    |
| Lactose                      | 5.0 g    |
| Water                        | 1 000 ml |

### **C-3.9.2 Preparation**

Suspend the components in 1 000 ml water. Heat if necessary to dissolve the medium completely. Sterilize at 121 °C for 15 min in an autoclave. Cool as quickly as possible after sterilization. Adjust the pH after sterilization to  $6.9 \pm 0.2$ .

## **C-3.10 Lactose Broth Medium**

### **C-3.10.1 Composition**

|                              |          |
|------------------------------|----------|
| Beef extract                 | 3.0 g    |
| Pancreatic digest of gelatin | 5.0 g    |
| Lactose                      | 5.0 g    |
| Water                        | 1 000 ml |

### **C-3.10.2 Preparation**

Suspend the components in water and heat if necessary to dissolve the medium completely. Sterilize at 121 °C for 15 min in an autoclave. Adjust the pH after sterilization to  $6.9 \pm 0.2$ .

## **C-3.11 Levine Eosin Methylene Blue Agar Medium**

### **C-3.11.1 Composition**

|                              |          |
|------------------------------|----------|
| Pancreatic digest of gelatin | 10.0 g   |
| Dibasic potassium phosphate  | 2.0 g    |
| Agar                         | 15.0 g   |
| Lactose                      | 10.0 g   |
| Eosin Y                      | 400 mg   |
| Methylene blue               | 65 mg    |
| Water                        | 1 000 ml |

### **C-3.11.2 Preparation**

Dissolve the pancreatic digest of gelatin, dibasic potassium phosphate and agar in water with warming and allow to cool. Just prior to use, liquify the gelled agar solution and the remaining

ingredients, as solutions, in the following amounts and mix. For each 100 ml of the liquified agar solution use 5 ml of a 20 percent w/v solution of lactose, 2 ml of a 2 percent w/v solution of eosin Y and 2 ml of a 0.33 percent w/v solution of methylene blue. The finished medium may not be clear. Adjust the pH after sterilization to  $7.1 \pm 0.2$ .

### C-3.12 MacConkey Agar Medium

#### C-3.12.1 Composition

|                                        |          |
|----------------------------------------|----------|
| Pancreatic digest of gelatin           | 17.0 g   |
| Peptone (meat and casein, equal parts) | 3.0 g    |
| Lactose                                | 10.0 g   |
| Sodium chloride                        | 5.0 g    |
| Bile salts                             | 1.5 g    |
| Agar                                   | 13.5 g   |
| Neutral red                            | 30 mg    |
| Crystal violet                         | 1 mg     |
| Water                                  | 1 000 ml |

#### C-3.12.2 Preparation

Boil the mixture of solids and water for 1 min to effect solution. Sterilize at 121 °C for 15 min in an autoclave. Adjust the pH after sterilization to  $7.1 \pm 0.2$ .

### C-3.13 MacConkey Broth Medium

#### C-3.13.1 Composition

|                              |          |
|------------------------------|----------|
| Pancreatic digest of gelatin | 20.0 g   |
| Lactose                      | 10.0 g   |
| Dehydrated ox bile           | 5.0 g    |
| Bromocresol purple           | 10 mg    |
| Water                        | 1 000 ml |

#### C-3.13.2 Preparation

Suspend the components in 1000 ml of water and heat if necessary to dissolve the medium completely. Sterilize at 121 °C for 15 min in an autoclave and adjust the pH after sterilization to  $7.3 \pm 0.2$ .

### C-3.14 Mannitol Salt Agar Medium

#### C-3.14.1 Composition

|                                |        |
|--------------------------------|--------|
| Pancreatic digest of gelatin   | 5.0 g  |
| Peptic digest of animal tissue | 5.0 g  |
| Beef extract                   | 1.0 g  |
| D-Mannitol                     | 10.0 g |
| Sodium chloride                | 75.0 g |
| Agar                           | 15.0 g |
| Phenol red                     | 25 mg  |

Water 1 000 ml

#### **C-3.14.2 Preparation**

Mix the components, heat with frequent agitation and boil for 1 min to effect solution. Sterilize at 121 °C for 15 min in an autoclave and adjust the pH after sterilization to  $7.4 \pm 0.2$ .

#### **C-3.15 Nutrient Broth Medium**

##### **C-3.15.1 Composition**

|                 |          |
|-----------------|----------|
| Beef extract    | 10.0 g   |
| Peptone         | 10.0 g   |
| Sodium chloride | 5 mg     |
| Water           | 1 000 ml |

##### **C-3.15.2 Preparation**

Dissolve the components with the aid of heat. Adjust the pH to 8.0 to 8.4 with 5M sodium hydroxide and boil for 10 min. Filter and sterilize by maintaining at 115 °C for 30 min and adjust pH to  $7.3 \pm 0.1$ .

#### **C-3.16 Nutrient Agar Medium**

Nutrient broth gelled by the addition of 1 to 2 percent w/v of agar.

#### **C-3.17 Pseudomonas Agar Medium for Detection of Flourescein**

##### **C-3.17.1 Composition**

|                                       |          |
|---------------------------------------|----------|
| Pancreatic digest of casein           | 10.0 g   |
| Peptic digest of animal tissue        | 10.0 g   |
| Anhydrous dibasic potassium phosphate | 1.5 g    |
| Magnesium sulphate hepta hydrate      | 1.5 g    |
| Glycerin                              | 10.0 ml  |
| Agar                                  | 15.0 g   |
| Water                                 | 1 000 ml |

##### **C-3.17.2 Preparation**

Dissolve the solid components in water before adding glycerin. Heat with frequent agitation and boil for 1 min to effect solution. Sterilize at 121 °C for 15 min in an autoclave and adjust the pH after sterilization to  $7.2 \pm 0.2$ .

#### **C-3.18 Pseudomonas Agar Medium for Detection of Pyocyanin**

##### **C-3.18.1 Composition**

|                              |        |
|------------------------------|--------|
| Pancreatic digest of gelatin | 20.0 g |
| Anhydrous magnesium chloride | 1.4 g  |

|                              |         |
|------------------------------|---------|
| Anhydrous potassium sulphate | 10.0 g  |
| Agar                         | 15.0 g  |
| Glycerin                     | 10.0 ml |
| Water                        | 1000 ml |

### **C-3.18.2 Preparation**

Dissolve the solid components in water before adding glycerin. Heat with frequent agitation and boil for 1 min to effect solution. Sterilize at 121 °C for 15 min in an autoclave and adjust the pH after sterilization to  $7.2 \pm 0.2$ .

## **C-3.19 Sabouraud Dextrose Agar Medium**

### **C-3.19.1 Composition**

|                                                                      |          |
|----------------------------------------------------------------------|----------|
| Dextrose                                                             | 40 g     |
| Peptic digest of animal tissue and Pancreatic digest of casein (1:1) | 10 g     |
| Agar                                                                 | 15 g     |
| Water                                                                | 1 000 ml |

### **C-3.19.2 Preparation**

Mix the components and heat to boiling to dissolve completely. Sterilize at 121 °C for 15 min in an autoclave and adjust the pH after sterilization to  $5.6 \pm 0.2$ .

## **C-3.20 Sabouraud Dextrose Agar Medium with Antibiotics**

To 1 000 ml of Sabouraud Dextrose Agar Medium, add 0.1 g of benzylpenicillin sodium and 0.1 g of tetracycline HCl or alternatively add 50 mg of chloramphenicol immediately before use.

## **C-3.21 Selenite F Broth**

### **C-3.21.1 Composition**

|                             |          |
|-----------------------------|----------|
| Peptone                     | 5 g      |
| Lactose                     | 4 g      |
| Disodium hydrogen phosphate | 10 g     |
| Sodium hydrogen selenite    | 4 g      |
| Water                       | 1 000 ml |

### **C-3.21.2 Preparation**

Suspend the components in water and mix well. Warm to dissolve the medium completely. Distribute in sterile containers and sterilize by maintaining at 100 °C for 30 min.

## **C-3.22 Fluid Selenite Cystine Medium**

### **C-3.22.1 Composition**

|                             |       |
|-----------------------------|-------|
| Pancreatic digest of casein | 5.0 g |
|-----------------------------|-------|

|                          |          |
|--------------------------|----------|
| Lactose                  | 4.0 g    |
| Sodium phosphate         | 10.0 g   |
| Sodium hydrogen selenite | 4.0 g    |
| L-Cystine                | 10.0 mg  |
| Water                    | 1 000 ml |

#### **C-3.22.2 Preparation**

Suspend the components in water and heat in flowing steam for 15 min. Adjust the final pH to  $7.0 \pm 0.2$ . Do not sterilize.

### **C-3.23 Tetrathionate Broth Medium**

#### **C-3.23.1 Composition**

|                     |          |
|---------------------|----------|
| Beef extract        | 0.9 g    |
| Peptone             | 4.5 g    |
| Yeast extract       | 1.8 g    |
| Sodium chloride     | 4.5 g    |
| Calcium carbonate   | 25.0 g   |
| Sodium thiosulphate | 40.7 g   |
| Water               | 1 000 ml |

#### **C-3.23.2 Preparation**

Dissolve the solids in 1 000 ml of water and heat the solution to boil. On the day of use, add a solution prepared by dissolving 5 g of potassium iodide and 6 g of iodine in 20 ml of water.

### **C-3.24 Tetrathionate Bile Brilliant Green Broth Medium**

#### **C-3.24.1 Composition**

|                         |          |
|-------------------------|----------|
| Peptone                 | 8.6 g    |
| Dehydrated ox bile      | 8.0 g    |
| Sodium chloride         | 6.4 g    |
| Calcium carbonate       | 20.0 g   |
| Potassium tetrathionate | 20.0 g   |
| Brilliant green         | 70 mg    |
| Water                   | 1 000 ml |

#### **C-3.24.2 Preparation**

Suspend the components in 1 000 ml of water. Heat just to boiling. Do not autoclave or reheat. Adjust the pH so that after heating it is  $7.0 \pm 0.2$ .

### **C-3.25 Triple Sugar Iron Agar Medium**

#### **C-3.25.1 Composition**

|              |       |
|--------------|-------|
| Beef extract | 3.0 g |
|--------------|-------|

|                      |          |
|----------------------|----------|
| Yeast extract        | 3.0 g    |
| Peptone              | 20.0 g   |
| Lactose              | 10.0 g   |
| Sucrose              | 10.0 g   |
| Dextrose monohydrate | 1.0 g    |
| Ferrous sulphate     | 0.2 g    |
| Sodium chloride      | 5.0 g    |
| Sodium thiosulphate  | 0.3 g    |
| Phenol red           | 24 mg    |
| Water                | 1 000 ml |

#### **C-3.25.2 Preparation**

Mix the components in 1 000 ml of water, allow standing for 15 min. Heat to boiling to dissolve the medium completely. Mix well and distribute into test tubes and Sterilize by maintaining at 121 °C for 15 min. Allow the medium to stand in a sloped form with a butt about 2.5 cm long.

#### **C-3.26 Urea Broth Medium**

##### **C-3.26.1 Composition**

|                                       |          |
|---------------------------------------|----------|
| Potassium dihydrogen orthophosphate   | 9.1 g    |
| Anhydrous disodium hydrogen phosphate | 9.5 g    |
| Urea                                  | 20.0 g   |
| Yeast extract                         | 0.1 g    |
| Phenol red                            | 10 mg    |
| Water                                 | 1 000 ml |

##### **C-3.26.2 Preparation**

Mix the components, sterilize by filtration and distribute aseptically in sterile containers.

#### **C-3.27 Vogel Johnson Agar Medium**

##### **C-3.27.1 Composition**

|                             |          |
|-----------------------------|----------|
| Pancreatic digest of casein | 10.0 g   |
| Yeast extract               | 5.0 g    |
| Mannitol                    | 10.0 g   |
| Dibasic potassium phosphate | 5.0 g    |
| Lithium chloride            | 5.0 g    |
| Glycerin                    | 10.0 g   |
| Agar                        | 16.0 g   |
| Phenol red                  | 25.0 mg  |
| Water                       | 1 000 ml |

##### **C-3.27.2 Preparation**

Suspend the components in 1 000 ml of water. Boil the solution of solids for 1 min. Sterilize at 121 °C for 15 min in an autoclave. Cool to 45 °C to 50 °C and add 20 ml of 1 percent w/v sterile solution of potassium tellurite. Adjust the pH after sterilization to  $7.0 \pm 0.2$ .

### C-3.28 Xylose Lysine Deoxycholate Agar Medium

#### C-3.28.1 Composition

|                         |          |
|-------------------------|----------|
| Xylose                  | 3.5 g    |
| L-Lysine                | 5.0 g    |
| Lactose                 | 7.5 g    |
| Sucrose                 | 7.5 g    |
| Sodium chloride         | 5.0 g    |
| Yeast extract           | 3.0 g    |
| Phenol red              | 80 mg    |
| Agar                    | 13.5 g   |
| Sodium deoxycholate     | 2.5 g    |
| Sodium thiosulphate     | 6.8 g    |
| Ferric ammonium citrate | 800 mg   |
| Water                   | 1 000 ml |

#### C-3.28.2 Preparation

Suspend the components in 1 000 ml of water. Heat with frequent agitation until the medium boils. Do not overheat or sterilize. Transfer at once to a water bath maintained at about 50 °C and pour into plates as soon as the medium has cooled. Adjust the final pH to  $7.4 \pm 0.2$ .

### C-4 SAMPLING

Use 10 ml or 10 g specimens for each of the tests specified in the Indian Standard.

### C-5 PRECAUTIONS

The microbial limit tests should be carried out under conditions designed to avoid accidental contamination during the test. The precautions taken to avoid contamination must be such that they do not adversely affect any microorganisms that should be revealed in the test.

### C-6 PRELIMINARY TESTING

Prior to doing the tests, inoculate diluted extracts being examined with separate viable cultures of *Escherichia coli*, *Salmonella* species, *Pseudomonas aeruginosa* and *Staphylococcus aureus*. This is done by adding 1 ml of 24 h broth culture containing not less than 1000 microorganisms to the first dilution (in buffer solution pH 7.2, fluid soyabean casein digest medium or fluid lactose medium) of the test material and following the test procedure. If the organisms fail to grow in the relevant medium the procedure should be modified by (a) increasing the volume of diluent with the quantity of test material remaining the same, or (b) incorporating a sufficient quantity of a suitable inactivating agent in the diluents, or (c) combining the afore mentioned modifications so as to permit growth of the organisms in the media. If inhibitory substances are present in the extracts, 0.5 percent of soya lecithin and 4 percent of polysorbate 20 may be added to the culture medium. Alternatively, repeat the test as described in the previous paragraph, using fluid casein digest soya lecithin polysorbate 20 medium, to demonstrate neutralization of preservatives or other antimicrobial agents in the test material. Where

inhibitory substances are contained in the extracts and the latter is soluble, the membrane filtration method described under Total Aerobic Microbial Count may be used. If in spite of incorporation of suitable inactivating agents and a substantial increase in the volume of diluent it is still not possible to recover the viable cultures described above and where the article is not suitable for applying the membrane filtration method, it can be assumed that the failure to isolate the inoculated organism may be due to the bactericidal activity of the product. This may indicate that the article is not likely to be contaminated with the given species of microorganisms. However, monitoring should be continued to establish the spectrum of inhibition and bactericidal activity of the article.

## **C-7 TOTAL AEROBIC MICROBIAL COUNT**

Pre-treat the extracts and raw materials being examined as described below.

NOTE – The raw materials need to be ground as a coarse powder before analysis.

### **C-7.1 Water Soluble Products**

Dissolve 10 g or dilute 10 ml of the extract preparation being examined, unless otherwise specified, in buffered sodium chloride peptone solution pH 7.0 or any other suitable medium shown to have no antimicrobial activity under the conditions of test and adjust the volume to 100 ml with the same medium. If necessary, adjust the pH to about 7.

### **C-7.2 Products Insoluble in Water (Non Fatty)**

Suspend 10 g or 10 ml of the extract preparation being examined, unless otherwise specified, in buffered sodium chloride peptone solution pH 7.0 or any other suitable medium shown not to have antimicrobial activity under the conditions of the test and dilute to 100 ml with the same medium. If necessary, divide the preparation being examined and homogenize the suspension mechanically. A suitable surface active agent such as 0.1 percent w/v of polysorbate 80 may be added to assist the suspension of poorly wettable substances. If necessary, adjust pH of the suspension to about 7.

### **C-7.3 Fatty Products**

Homogenise 10 g or 10 ml of the extract preparation being examined, unless otherwise specified, with 5 g of polysorbate 20 or polysorbate 80. If necessary, heat to not more than 40 °C. Mix carefully while maintaining the temperature in the water bath or in an oven. Add 85 ml of buffered sodium chloride peptone solution pH 7.0 or any other suitable medium shown to have no antimicrobial activity under the conditions of the test, heated to not more than 40 °C if necessary. Maintain this temperature for the shortest time necessary for formation of an emulsion and in any case for not more than 30 min. If necessary, adjust the pH to about 7.

### **C-7.4 Examination of the Sample**

Determine the total aerobic microbial count in the extract being examined by any of the following methods.

### **C-7.5 Membrane Filtration**

Use membrane filters 50 mm in diameter and having a nominal pore size not greater than 0.45 µm the effectiveness of which in retaining bacteria has been established for the type of preparation being

examined. Transfer 10 ml or a quantity of each dilution containing 1g of the preparation being examined to each of two membrane filters and filter immediately. If necessary, dilute the pretreated extract preparation so that a colony count of 10 to 100 may be expected. Wash each membrane by filtering through it three or more successive quantities, each of about 100 ml, of a suitable liquid such as buffered sodium chloride peptone solution pH 7.0. For fatty substances add to the liquid polysorbate 20 or polysorbate 80. Transfer one of the membrane filters, intended for the enumeration of bacteria, to the surface of a plate of casein soyabean digest agar and the other, intended for the enumeration of fungi, to the surface of a plate of Sabouraud dextrose agar with antibiotics.

Incubate the plates for 5 days, unless a more reliable count is obtained in shorter time, at 30 °C to 35 °C in the test for bacteria and 20 °C to 25 °C in the test for fungi. Count the number of colonies that are formed. Calculate the number of microorganisms per g or per ml of the extract preparation being examined, if necessary, count bacteria and fungi separately.

#### **C-7.6 Plate Count for Bacteria**

Using Petri dishes 9 cm to 10 cm in diameter, add to each dish a mixture of 1 ml of the pretreated extract preparation and about 15 ml of liquified casein soyabean digest agar at not more than 45 °C.

Alternatively, spread the pretreated extract preparation on the surface of the solidified medium in a petri dish of the same diameter. If necessary, dilute the pretreated extract preparation as described above so that a colony count of not more than 300 may be expected. Prepare at least two such petri dishes using the same dilution and incubate at 30 °C to 35 °C for 5 days, unless a more reliable count is obtained in a shorter time. Count the number of colonies that are formed. Calculate the results using plates with the greatest number of colonies but taking 300 colonies per plate as the maximum consistent with good evaluation.

#### **C-7.7 Plate Count for Fungi**

Proceed as described in the test for bacteria but use Sabouraud dextrose agar with antibiotics in place of casein soyabean digest agar and incubate the plates at 20 °C to 25 °C for 5 days, unless a more reliable count is obtained in a shorter time. Calculate the results using plates with not more than 100 colonies.

#### **C-7.8 Multiple Tube or Serial Dilution Method**

In each of fourteen test tubes of similar size place 9.0 ml of sterile fluid soyabean casein digest medium. Arrange twelve of the tubes in four sets of three tubes each. Put aside one set of three tubes to serve as controls. Into each of three tubes of one set ("100") and into fourth tube (A) pipette 1 ml of the solution of suspension of the test specimen (extract) and mix. From tube A pipette 1 ml of its contents into the one remaining tube (B) not included in the set and mix. These two tubes contain 100 mg (or 100 µl) and 10 mg (or 10 µl) of the specimen respectively. Into each of the second set ("10") of three tubes pipette 1 ml from tube A, and into each tube of the third set ("1") pipette 1 ml from tube B. Discard the unused contents of tube A and B. Close well and incubate all of the tubes. Following the incubation period, examine the tubes for growth. The three control tubes remain clear. Observations in the tubes containing the test specimen, when interpreted by reference to Table 2, indicate the most probable number of microorganisms per g or per ml of the test specimen.

**Table 2 Most probable Total Count by Multiple Tube or Serial Dilution Method**

(Clause C-7.8)

| Sl No | Observed Combination of Numbers of Tubes<br>Showing Growth in Each Set |         |        | Most Probable<br>Number of<br>Microorganisms<br>per g or per ml |
|-------|------------------------------------------------------------------------|---------|--------|-----------------------------------------------------------------|
|       | Number of mg (or ml) of Specimen per Tube                              |         |        |                                                                 |
|       | 100                                                                    | 10      | 1      |                                                                 |
|       | (100 µL)                                                               | (10 µL) | (1 µL) |                                                                 |
| (1)   | (2)                                                                    | (3)     | (4)    | (5)                                                             |
| i)    | 3                                                                      | 3       | 3      | >1100                                                           |
| ii)   | 3                                                                      | 3       | 2      | 1100                                                            |
| iii)  | 3                                                                      | 3       | 1      | 500                                                             |
| iv)   | 3                                                                      | 3       | 0      | 200                                                             |
| v)    | 3                                                                      | 2       | 3      | 290                                                             |
| vi)   | 3                                                                      | 2       | 2      | 210                                                             |
| vii)  | 3                                                                      | 2       | 1      | 150                                                             |
| viii) | 3                                                                      | 2       | 0      | 90                                                              |
| ix)   | 3                                                                      | 1       | 3      | 160                                                             |
| x)    | 3                                                                      | 1       | 2      | 120                                                             |
| xi)   | 3                                                                      | 1       | 1      | 70                                                              |
| xii)  | 3                                                                      | 1       | 0      | 40                                                              |
| xiii) | 3                                                                      | 0       | 3      | 95                                                              |
| xiv)  | 3                                                                      | 0       | 2      | 60                                                              |
| xv)   | 3                                                                      | 0       | 1      | 40                                                              |
| xvi)  | 3                                                                      | 0       | 0      | 23                                                              |

## C-8 TESTS FOR SPECIFIED MICROORGANISMS

### C-8.1 Pre-treatment of the Extract Sample Being Examined

Proceed as described under the test for total aerobic microbial count but using lactose broth or any other suitable medium shown to have no antimicrobial activity under the conditions of test in place of buffered sodium chloride peptone solution pH 7.0.

### C-8.2 *Escherichia coli*

Place the prescribed quantity in a sterile screw capped container, add 50 ml of nutrient broth, shake, allow to stand for 1 h (4 h for gelatin) and shake again. Loosen the cap and incubate at 37 °C for 18 h to 24 h.

#### C-8.2.1 Primary Test

Add 1.0 ml of the enrichment culture to a tube containing 5 ml of MacConkey broth. Incubate in a water bath at 36 °C to 38 °C for 48 h. If the contents of the tube show acid and gas, carry out the secondary test.

#### C-8.2.2 Secondary Test

Add 0.1 ml of the contents of the tubes containing (a) 5 ml of MacConkey broth and (b) 5 ml of peptone water. Incubate in a water bath at 43.5 °C to 44.5 °C for 24 h and examine tube (a) for acid and gas and tube (b) for indole. To test for indole, add 0.5 ml of Kovac's reagent, shake well and allow to stand for 1 min. If a red colour is produced in the reagent layer indole is present. The presence of acid and gas and of indole in the secondary test indicates the presence of *Escherichia coli*.

Carry out a control test by repeating the primary and secondary tests, adding 1.0 ml of the enrichment culture and a volume of broth containing 10 to 50 *Escherichia coli* (NCTC 9002) organisms, prepared from a 24 h culture in nutrient broth, to 5 ml of MacConkey broth. The test is not valid unless the results indicate that the control contains *Escherichia coli*.

### C-8.2.3 Alternative Test

By means of an inoculating loop, streak a portion from the enrichment culture (obtained in the previous test) on the surface of MacConkey agar medium. Cover and invert the dishes and incubate.

Upon examination, if none of the colonies are brick red in colour and have a surrounding zone of precipitated bile the sample meets the requirements of the test for the absence of *Escherichia coli*. If the colonies described above are found, transfer the suspect colonies individually to the surface of Levine eosin methylene blue agar medium, plated on petri dishes. Cover and invert the plates and incubate. Upon examination, if none of the colonies exhibits both a characteristic metallic sheen under reflected light and a blue-black appearance under transmitted light, the sample meets the requirements of the test for the absence of *Escherichia coli*. The presence of *Escherichia coli* may be confirmed by further suitable cultural and biochemical tests.

## C-8.3 Salmonella

Transfer a quantity of the pretreated preparation being examined containing 1 g or 1 ml of the product to 100 ml of nutrient broth in a sterile screw capped jar, shake, allow to stand for 4 h and shake again. Loosen the cap and incubate at 35 °C to 37 °C for 24 h.

### C-8.3.1 Primary Test

Add 1.0 ml of the enrichment culture to each of the two tubes containing (a) 10 ml of selenite F broth and (b) tetrathionate bile brilliant green broth and incubate at 36 °C to 38 °C for 48 h. From each of these two cultures subculture on at least two of the following four agar media: bismuth sulphate agar, brilliant green agar, deoxycholate citrate agar and xylose lysine deoxycholate agar. Incubate the plates at 36 °C to 38 °C for 18 to 24 h. Upon examination, if none of the colonies conforms to the description given in Table 3, the sample meets the requirements of the test for the absence of the genus *Salmonella*. If any colonies conforming to the description in Table 3 are produced, carry out the secondary test.

**Table 3 Interpretation of the Confirmatory Tests for *Salmonella***

(Clause C-8.3.1)

| Sl No<br>(1) | Medium<br>(2)         | Description of Colony<br>(3)                                                                                |
|--------------|-----------------------|-------------------------------------------------------------------------------------------------------------|
| i)           | Bismuth sulphite agar | Black or green                                                                                              |
| ii)          | Brilliant green agar  | Small, transparent and colorless, or opaque, pinkish or white (frequently surrounded by a pink or red zone) |

|      |                                 |                                                     |
|------|---------------------------------|-----------------------------------------------------|
| iii) | Deoxycholate citrate agar       | Colorless and opaque, with or without black centers |
| iv)  | Xylose lysine deoxycholate agar | Red with or without black centres                   |

### C-8.3.2 Secondary Test

Subculture any colonies showing the characteristics given in Table 3 in triple sugar iron agar by first inoculating the surface of the slope and then making a stab culture with the same inoculating needle, and at the same time inoculate a tube of urea broth. Incubate at 36 °C to 38 °C for 18 h to 24 h. The formation of acid and gas in the stab culture (with or without concomitant blackening) and the absence of acidity from the surface growth in the triple sugar iron agar, together with the absence of a red color in the urea broth, indicate the presence of *Salmonella*. If acid but no gas is produced in the stab culture, the identity of the organisms should be confirmed by agglutination tests.

Carry out the control test by repeating the primary and secondary tests using 1.0 ml of the enrichment culture and a volume of broth containing 10 to 50 *Salmonella abony* (NCTC 6017) organisms, prepared from a 24 h culture in nutrient broth, for the inoculation of the tubes (a) and (b). The test is not valid unless the results indicate that the control contains *Salmonella*.

### C-8.4 *Pseudomonas aeruginosa*

Pretreat the preparation being examined as described above and inoculate 100 ml of fluid soyabean casein digest medium with a quantity of the solution, suspension or emulsion thus obtained containing 1 g or 1 ml of the preparation being examined. Mix and incubate at 35 °C to 37 °C for 24 h to 48 h. Examine the medium for growth and if growth is present, streak a portion of the medium on the surface of cetrimide agar medium, each plated on petri dishes. Cover and incubate at 35 °C to 37 °C for 18 h to 24 h.

If, upon examination, none of the plates contains colonies having the characteristics listed in Table 4 for the media used, the sample meets the requirement for freedom from *Pseudomonas aeruginosa*. If any colonies conforming to the description in Table 4 are produced, carry out the oxidase and pigment tests. Streak representative suspect colonies from the agar surface of cetrimide agar on the surfaces of *Pseudomonas* agar medium for detection of fluorescein and *Pseudomonas* agar medium for detection of pyocyanin contained in petri dishes. Cover and invert the inoculated media and incubate at 33 °C to 37 °C for not less than 3 days. Examine the streaked surfaces under ultra violet light. Examine the plates to determine whether colonies conforming to the description in Table 4 are present. If growth of suspect colonies occurs, place 2 or 3 drops of a freshly prepared 1 percent w/v solution of N,N,N1,N1-tetramethyl-4-phenylenediamine dihydrochloride on filter paper and smear with the colony. If there is no development of a pink color, changing to purple, the sample meets the requirements of the test for the absence of *Pseudomonas aeruginosa*.

**Table 4 Interpretation of the Confirmatory Tests for *Pseudomonas aeruginosa***  
(Clause C -8.4)

| Sl No | Selective Medium | Characteristic<br>Colonial<br>Morphology | Fluorescence<br>in UV Light | Oxidase<br>Test | Gram Stain    |
|-------|------------------|------------------------------------------|-----------------------------|-----------------|---------------|
| (1)   | (2)              | (3)                                      | (4)                         | (5)             | (6)           |
| i)    | Cetrimide agar   | Generally<br>greenish                    | Greenish                    | Positive        | Negative rods |

|      |                                                      |                                  |           |          |               |
|------|------------------------------------------------------|----------------------------------|-----------|----------|---------------|
| ii)  | Pseudomonas agar medium for detection of fluorescein | Generally colorless to yellowish | Yellowish | Positive | Negative rods |
| iii) | Pseudomonas agar for detection of pyocyanin          | Generally greenish               | Blue      | Positive | Negative rods |

### C-8.5 *Staphylococcus aureus*

Proceed as described under *Pseudomonas aeruginosa*, if upon examination of the incubated plates, none of them contains colonies having the characteristics listed in the media used, the sample meets the requirements for the absence of *Staphylococcus aureus*. If growth occurs, carry out the coagulase test. Transfer representative suspect colonies from the agar surface of any of the media listed in Table 5 to individual tubes, each containing 0.5 ml of mammalian, preferably rabbit or horse, plasma with or without additives.

**Table 5 Interpretation of the Confirmatory Tests for *Staphylococcus aureus***  
(Clause C -8.5)

| Sl No.<br>(1) | Selective Medium<br>(2) | Characteristic Colonial Morphology<br>(3)               | Gram Stain<br>(4)            |
|---------------|-------------------------|---------------------------------------------------------|------------------------------|
| i)            | Vogel johnson agar      | Black surrounded by yellow zones                        | Positive cocci (in clusters) |
| ii)           | Mannitol salt agar      | Yellow colonies with yellow zones                       | Positive cocci (in clusters) |
| iii)          | Baird parker agar       | Black, shiny, surrounded by clear zones of 2 mm to 5 mm | Positive cocci (in clusters) |

Incubate in water bath at 37 °C examining the tubes at 3 h and subsequently at suitable intervals up to 24 h. If no coagulation in any degree is observed, the sample meets the requirements of the test for the absence of *Staphylococcus aureus*.

### C-8.6 Validity of the Tests for Total Aerobic Microbial Count

Grow the following test strains separately in tubes containing fluid soyabean casein digest medium at 30 °C to 35 °C for 18 h to 24 h or, for *Candida albicans*, at 20 °C for 48 h.

|                              |                         |
|------------------------------|-------------------------|
| <i>Staphylococcus aureus</i> | (ATCC 6538; NCTC 10788) |
| <i>Bacillus subtilis</i>     | (ATCC 6633; NCIB 8054)  |
| <i>Escherichia coli</i>      | (ATCC 8739; NCIB 8545)  |
| <i>Candida albicans</i>      | (ATCC 2091; ATCC 10231) |

Dilute portions of each of the cultures using buffered sodium chloride peptone solution pH 7.0 to make test suspensions containing about 100 viable microorganisms per ml. Use the suspension of each of the microorganisms separately as a control of the counting methods, in the presence and absence of the preparation being examined, if necessary.

A count for any of the test organisms differing by not more than a factor of 10 from the calculated value for the inoculum should be obtained. To test the sterility of the medium and of the diluent and

the aseptic performance of the test, carry out the total aerobic microbial count method using sterile buffered sodium chloride peptone solution pH 7.0 as the test preparation. There should be no growth of microorganisms.

#### **C-8.7 Validity of the Tests for Specified Microorganisms**

Grow separately the test strains of *Staphylococcus aureus* and *Pseudomonas aeruginosa* in fluid soyabean casein digest medium and *Escherichia coli* and *Salmonella typhimurium* at 30 °C to 35 °C for 18 h to 24 h. Dilute portions of each of the cultures using buffered sodium chloride peptone solution pH 7.0 to make test suspensions containing about  $10^3$  viable microorganisms per ml. Mix equal volume of each suspension and use 0.4 ml (approximately  $10^2$  microorganisms of each strain) as an inoculum in the test for *E. coli*, *S. typhimurium*, *P. aeruginosa* and *S. aureus*, in the presence and absence of the extract preparation being examined, if necessary. A positive result for the respective strain of microorganism should be obtained.